

Credit Risk Analysis & SQD: Application-Driven Workflow Patterns for HPC-QC System Integration



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Agenda

1. Credit Risk Management
2. Sample-based Quantum Diagonalization (SQD)
3. Comparative analysis
4. Open for discussion

Credit Risk Analysis

Credit Risk Management: Application & Algorithm



CRM is a critical function in financial services: assessing, monitoring, and mitigating potential losses arising from borrowers' failure to meet their financial obligations



Given a portfolio of K correlated assets, estimate the probability distribution of portfolio losses and derive risk measures (Expected Loss, VaR, CVaR)



High-dimensional, computationally intensive, a natural candidate for quantum advantage

Credit Risk Analysis: Classical vs. Quantum Methods

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Monte-Carlo samples millions of default/loss scenarios to build the distribution.

Achieving sub-percent VaR error typically requires overnight compute cycles.

Convergence rate at $O(1/M^2)$.



Encode the loss distribution into a quantum state using a Gaussian Conditional Independence (GCI) model.

Use Quantum Amplitude Estimation (QAE)[1] to extract risk measures directly from quantum amplitudes.

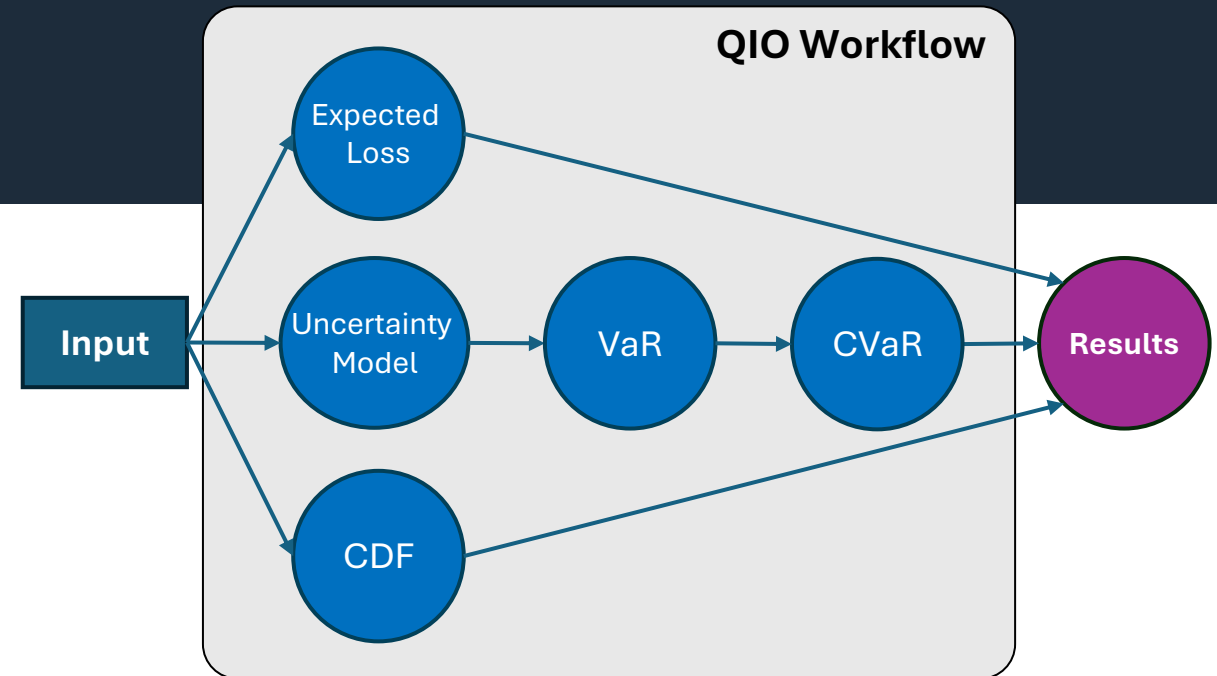
A single quantum run estimates the tail probability with $O(1/M)$ scaling.

Credit Risk Analysis: Orchestrable Task

The application is divided into distinct computational tasks :

- **Uncertainty Model** → quantum circuits encoding default probability distributions
- **Expected Loss** → average loss across all scenarios
- **Cumulative Distribution Function(CDF)** → probability of loss not exceeding a threshold
- **Value at Risk(VaR)** → loss level not exceeded with probability α
- **Conditional Value at Risk(CVaR)** → expected loss given that loss exceeds VaR

Key insight: some tasks are independent, others have dependencies, this structure determines what can be parallelized.



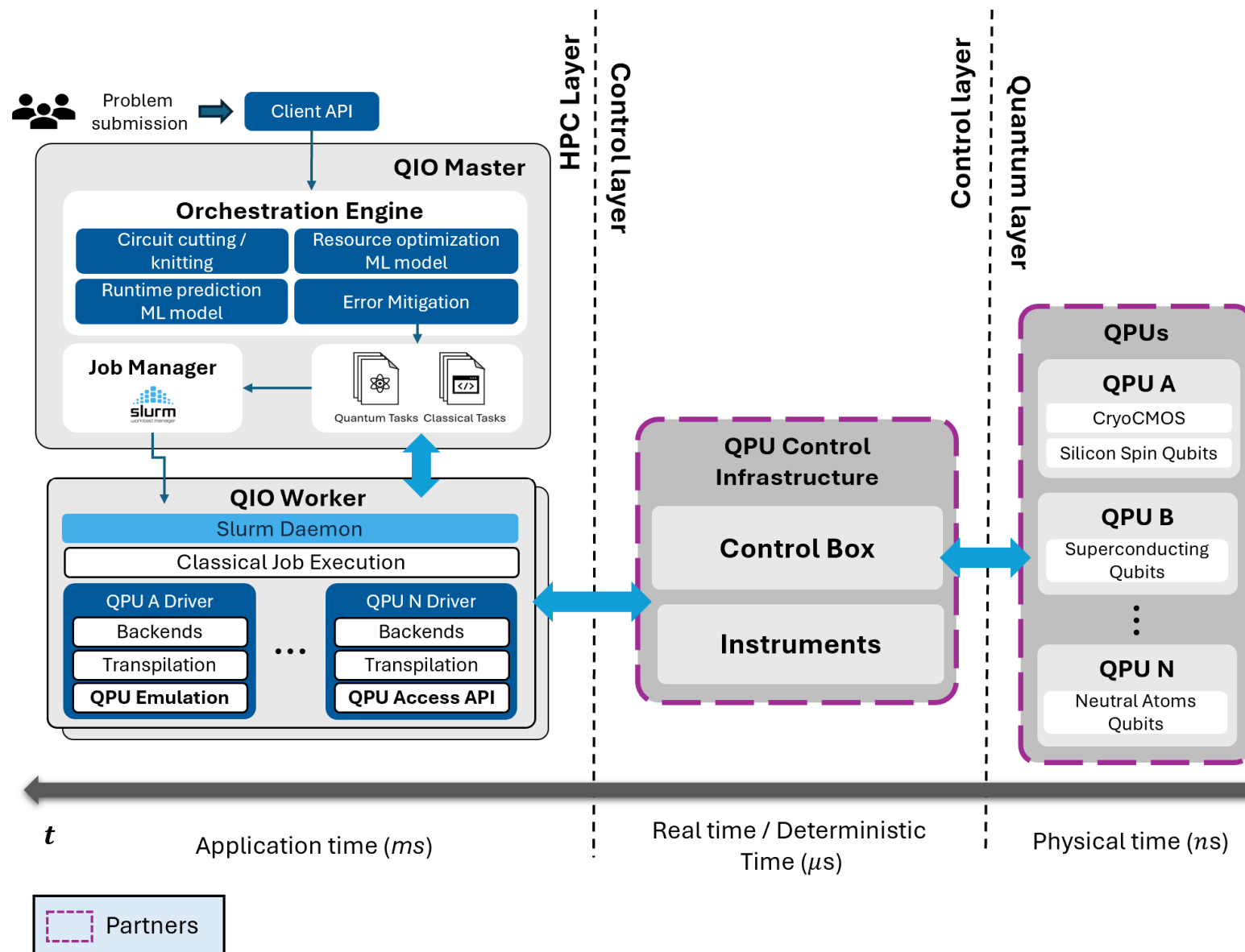
Credit Risk Analysis: Classical vs. Quantum Components

Workflow pattern and communication events:

1. **Classical** → **Classical**: Data ingestion, portfolio construction, individual default probabilities computed for K assets
2. **Classical** → **Quantum**: Uncertainty model circuit construction; circuit execution isolated via a QIO wrapper that decouples it from the classical workflow and dispatches to any available device
3. **Quantum** → **Classical**: Sampling/estimation results returned; classical tasks consume quantum outputs for loss aggregation
4. **Classical** → **Classical**: Risk measure computation (VaR, CVaR), regulatory reporting

Loosely coupled, application-time regime, communication is batch-style – no tight inner loop, but orchestration must respect the task dependency graph

Dell's Quantum Intelligent Orchestrator (QIO)



Credit Risk Analysis: Infrastructure & Scaling

Management layer: 2 VMs (8 GB RAM, 8 cores each) – SLURM control + orchestration service

CPU partition: 2 worker VMs (64 GB RAM, 20 cores each) – classical pre/post-processing

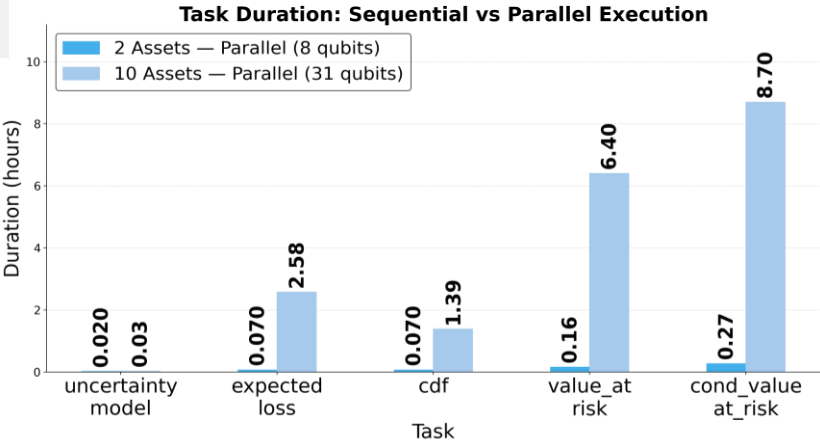
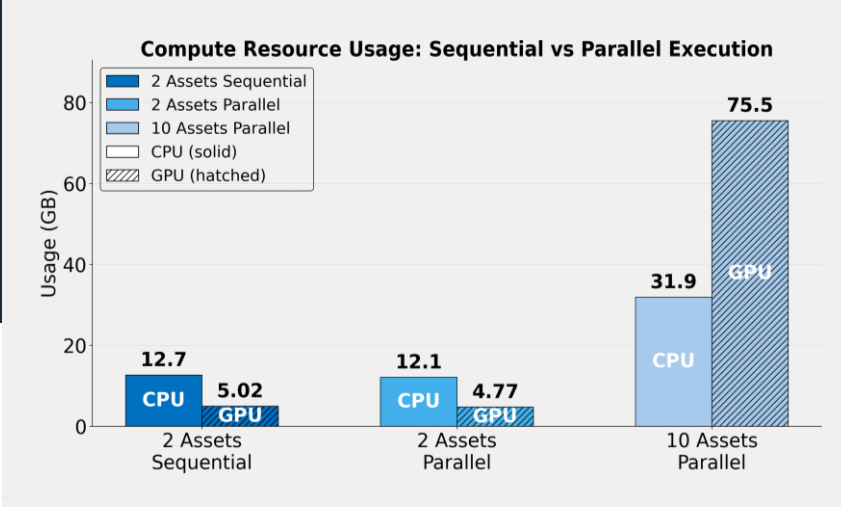
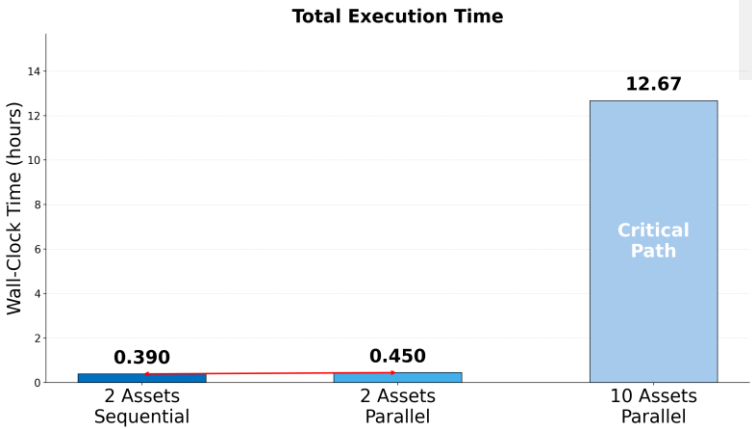
GPU partition: 1 bare-metal node (1 TB RAM, 128 cores, 4× NVIDIA L40S 40 GB) – quantum circuit simulation

Scaling pressure: 10 correlated assets → 31 qubits → GPU memory pushed to 75.5 GB, validating the need for dynamic resource allocation across heterogeneous partitions

Serial execution at this scale was not feasible; parallel orchestration is not optional, it's a requirement



Credit Risk Analysis: Results



Sample-based Quantum Diagonalization (SQD)

QSCI/SQD

?

QSCI

Quantum Selected
Configuration Interaction

=

?

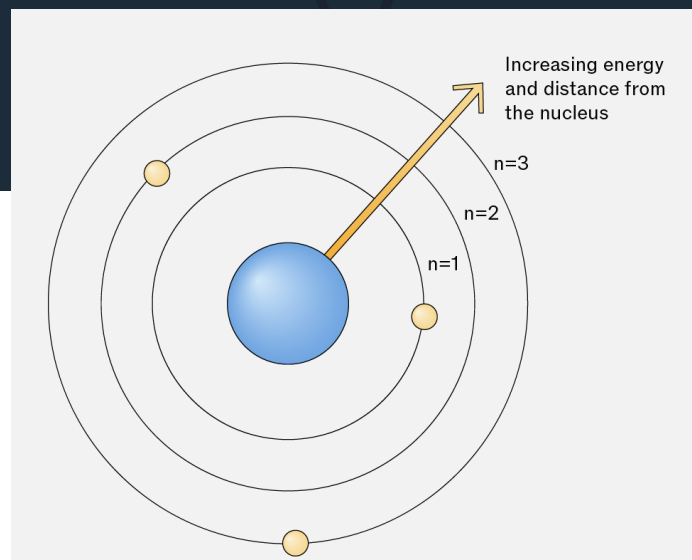
SQD

Sample-based Quantum
Diagonalization[3]

QSCI: Application & Algorithm



Computing ground-state energies of molecular systems is a central problem in quantum chemistry and one of the most promising candidates for quantum advantage



Enables analysis of large Hamiltonians that are intractable classically, and is robust to noise; works with imperfect QPU samples to produce meaningful results

QSCI: Classical vs. Quantum Methods

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Classical equivalent is exact diagonalization where we diagonalize the full Hamiltonian matrix — exponential scaling, infeasible for large molecular systems

Even approximate classical methods (CASCI, DMRG) struggle with strongly correlated systems

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Hybrid, uses a QPU to sample bitstrings from a candidate quantum state (the quantum device acts as a sampler, not a solver)

$$H_S^{(k)} = P_{S^{(k)}} H P_{S^{(k)}}$$

Classical HPC performs the heavy lifting: subspace construction, projection, and diagonalization in a reduced space

QSCI: Classical/HPC-Quantum Workflow

Workflow pattern and communication events:

1. **Classical** → **Quantum**: Ansatz circuit construction, transpilation, submission to QPU/simulator for sampling
2. **Quantum** → **Classical**: Bitstring samples returned, this is the primary quantum-to-classical communication event
3. **Classical** → **Classical**: Configuration recovery (filtering), subspace construction, projected Hamiltonian computation

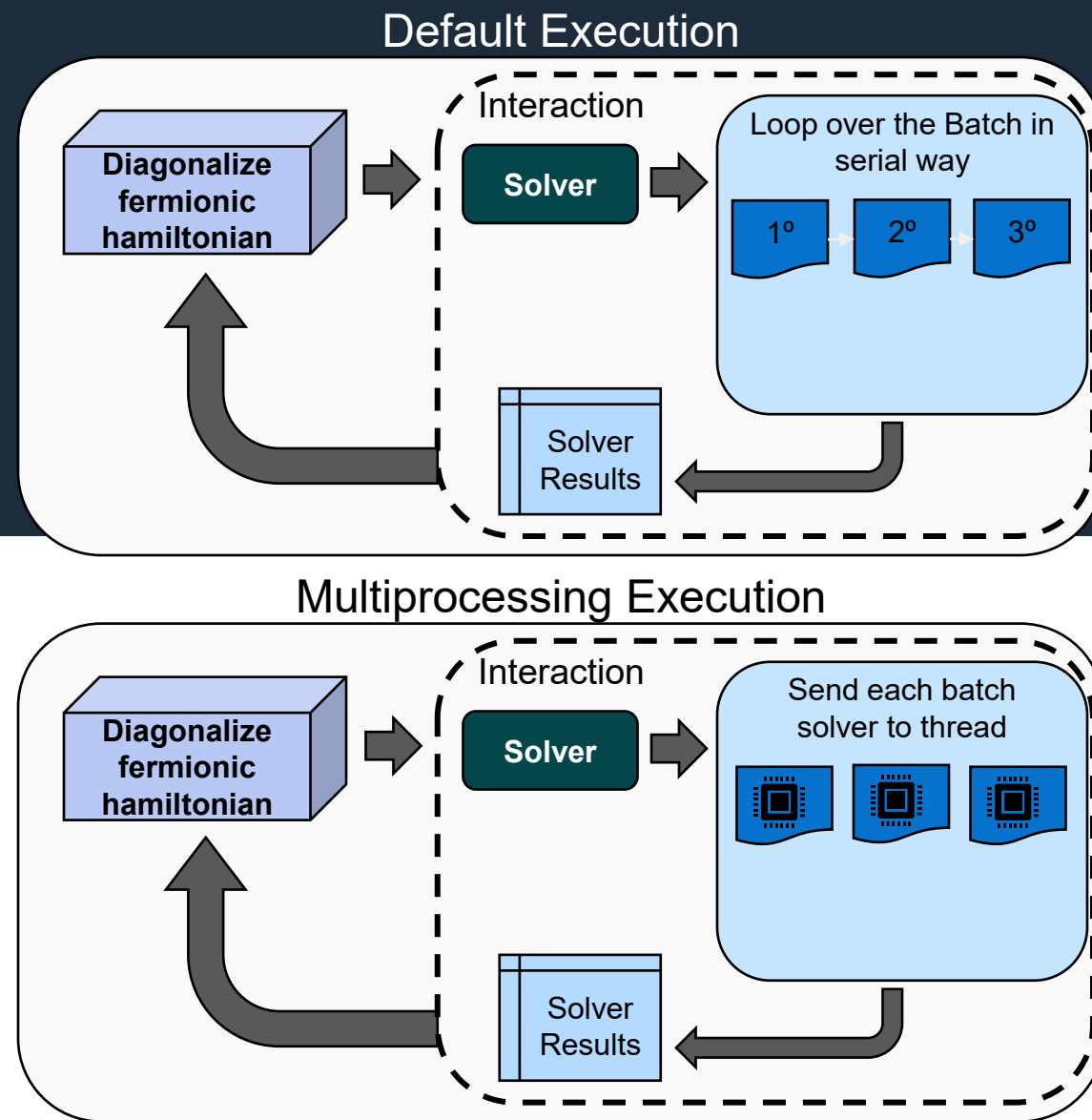
4. **Classical (HPC)**: Diagonalization

5. **Classical** → **Quantum (iterate)**: If not converged, update state preparation and resample

Loosely coupled, application-time regime, but with an iterative loop structure. Communication is sample-based meaning that the quantum device returns bitstrings, not expectation values.

QSCI: Orchestrable Task

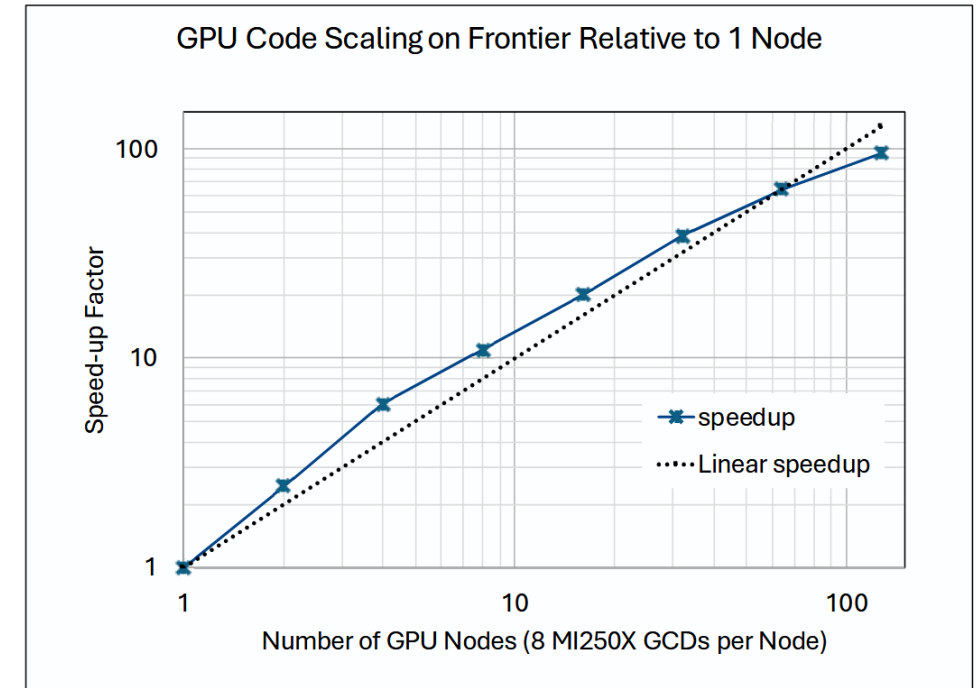
- Subspace diagonalization batches are independent, naturally parallelizable
- Each batch solver distributed across available cores via multiprocessing
- No modification to the SQD logic, same input/output interface preserved
- Direct speedup on the most compute-intensive workflow stage
- Validates the architectural path toward full orchestration via QIO



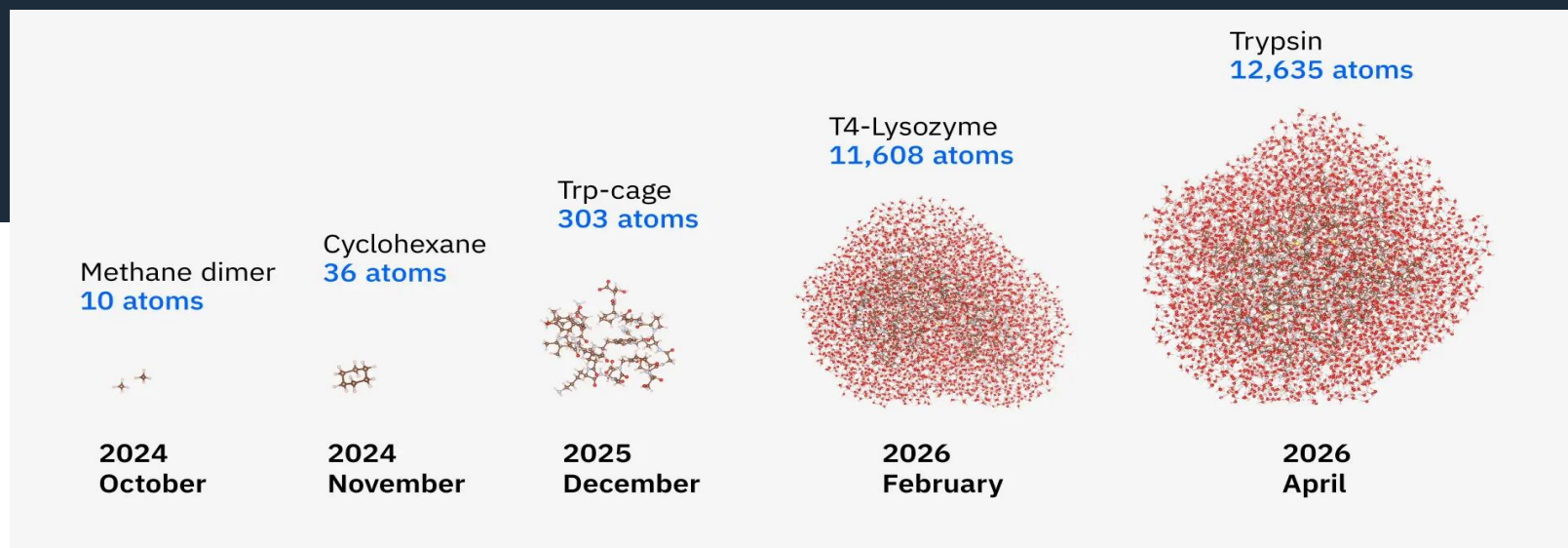
Relevant work in SQD

GPU-Accelerated Systems for SQD

- Ported SQD to GPU accelerators using OpenMP target offload directives, achieving 95× speedup on the Frontier supercomputer compared to CPU implementations.
- An additional 3× speedup was measured using the more recent MI300X GPU systems
- Transforming nested C++ data structures into flattened GPU-friendly array

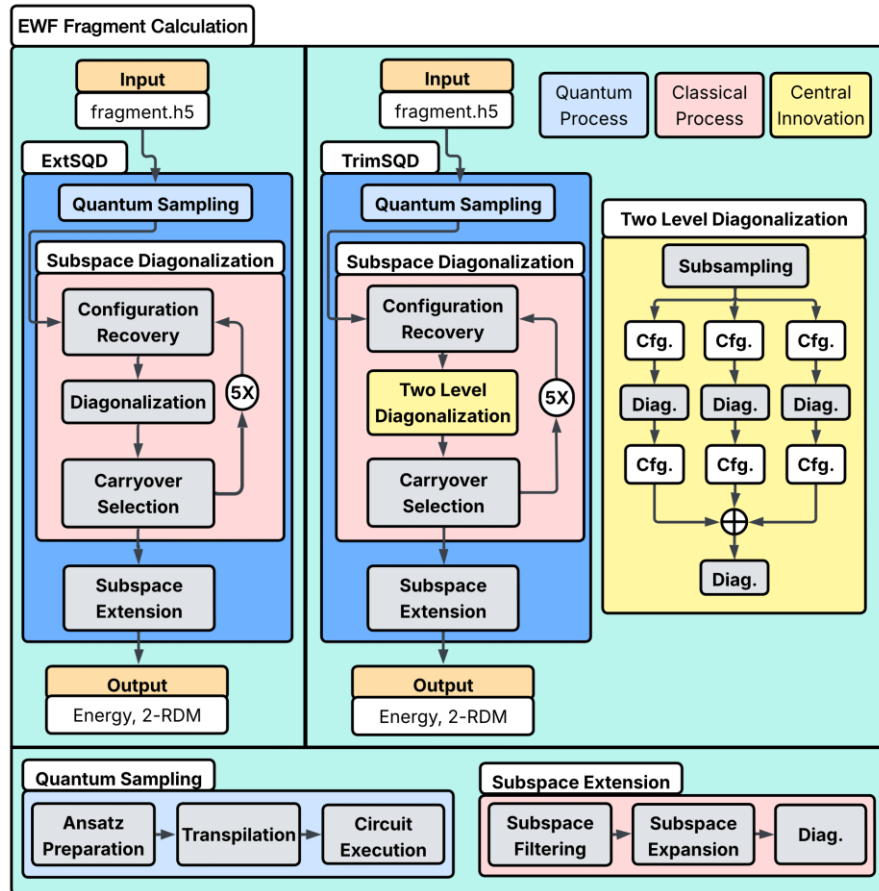


EWF + TrimSQD



Merz, K. M., Jr., Shajan, et al. (2026). Crossing the 12,000-atom barrier with heterogeneous quantum-classical supercomputing: quantum chemistry of protein-ligand complexes. preprint arXiv:2605.01138

EWF + TrimSQR



EWF fragments molecules into localized $< 10\text{\AA}$ clusters

The Win: TrimSQR filters out noise ("deadwood") instantly.

System Impact: Massively distributed hybrid scaling (QPU + Supercomputers)



SKQD

Samples across a dynamic
chain of time-evolved states

The Win: Provable
mathematical convergence
guarantees

System Impact: Deep circuits
require extreme QPU
coherence

Yu, J., Moreno, et al. (2025). Quantum-Centric Algorithm for Sample-Based Krylov Diagonalization. preprint arXiv:2501.09702

Compare CRM and SQD

Side-by-Side Comparison

Credit-risk analysis and SQD share a loosely-coupled, application-time execution model but stress different parts of the system.

	CRM	SQD
Domain	Finance	Quantum chemistry
Timing regime	Application-time	Application-time
Coupling	Loose, batch	Loose, iterative
Quantum role	Amplitude estimation	State sampling
Classical role	Aggregation, reporting	Heavy diagonalization (HPC)
Bottleneck	Orchestration, data gravity	Classical post-processing
Unique constraint	Governance, regulation	Compute intensity, noise robustness

References

1. [Quantum Risk Analysis. Stefan Woerner, Daniel J. Egger.](#) [Woerner2019]
2. [Credit Risk Analysis using Quantum Computers. Egger et al. \(2019\)](#) [Egger2019]
3. [Sample-based quantum diagonalization, IBM](#)

SQD: Circuit Size per Molecule grouped by Basis

